L3 SRE / DevOps — Real-Time Scenario Interview Q&A;

20 production-focused scenarios with interview-ready spoken answers

Question

Suppose a critical production application goes down at 3 AM and nobody from the application team is available. What will you do?

Spoken Answer

"First, I would acknowledge the incident and quickly establish the impact and scope—whether it's a complete outage, a particular region, or a specific functionality. I would check the monitoring dashboards, alerts, application logs, infrastructure metrics and traces to identify what changed or where the failure is occurring. If there is an established runbook, I would follow it. If the runbook doesn't cover the situation, I would focus first on safe mitigation, such as rollback, restarting a failed component, scaling, traffic diversion or failover, depending on the architecture. At the same time, I would communicate the business impact and current status through the incident channel and engage the appropriate on-call or escalation team. Once service is restored, I would validate the SLIs, document the timeline, perform RCA and update the runbook so the same situation can be handled faster next time." ---

2. You have three incidents at the same time

Question

You have a production outage, a performance issue reported by development, and a dashboard request. What do you prioritise?

Spoken Answer

"I prioritise based on customer and business impact. The production outage comes first because it directly affects availability. I would acknowledge the other requests so the stakeholders know they're being tracked. For the production issue, I would either handle it myself or delegate investigation depending on the team availability. Once the critical incident is stabilised, I would address the performance issue and then the dashboard request. I also make sure prioritisation is communicated clearly, because the important thing is not just solving the highest-priority issue but making sure stakeholders understand why other work is temporarily delayed." ---

3. Production deployment caused errors

Question

A deployment completed successfully, but five minutes later the error rate increased. What do you do?

Spoken Answer

"I would first correlate the timing of the deployment with the increase in errors. I would check the application error rate, latency, traffic and affected endpoints, then compare the current version with the previous version. I would use logs and traces to identify the actual failure and check whether a dependency such as the database, API or messaging system is also involved. If there is clear evidence that the deployment caused the issue and rollback is safe, I would initiate rollback according to the change procedure. After recovery, I would validate the customer-facing SLIs, identify the root cause and introduce preventive controls such as better automated testing, deployment gates or canary monitoring." ---

4. API latency suddenly increases

Question

Your API P99 latency increases from 500 ms to 5 seconds. What is your approach?

Spoken Answer

"I would first determine the scope—whether it's all APIs, one endpoint, one region or a particular customer journey. Then I would check request rate, error rate and P95/P99 latency in Grafana or the monitoring system. Next I would use distributed tracing to identify which span is consuming the additional time. If the application itself looks healthy, I would investigate downstream dependencies such as the database, external API, cache or messaging system. I would correlate the trace ID with application logs and infrastructure metrics. Once I identify the bottleneck, I would mitigate it—for example by scaling, rolling back a recent change, failing over or engaging the dependency team—and then validate that latency has returned within the SLO." ---

5. Database is intermittently slow

Question

Application is healthy most of the time, but database calls occasionally take 10 seconds. What do you do?

Spoken Answer

"I would first establish whether the latency is actually concentrated in database calls by looking at traces and application metrics. Then I would correlate the slow requests with database metrics such as connection utilisation, CPU, memory, I/O, locks, query latency and connection pool saturation. If I have the required database access, I would investigate the relevant database metrics and slow queries. If database administration is owned by another team, I would provide them the exact timestamps, query or service information, trace IDs and impact rather than simply saying the database is slow. In parallel, I would look for application-side issues such as connection pool exhaustion or inefficient retries." ---

6. Prometheus suddenly stops receiving metrics

Question

Prometheus is running, but suddenly there are no application metrics. What do you check?

Spoken Answer

"I would first determine whether the problem is with Prometheus itself or with a particular target. I would check the Prometheus targets page and look for scrape errors. Then I would verify that the application's metrics endpoint is available, usually the `/metrics` endpoint. In Kubernetes I would check the Service, endpoints, ServiceMonitor or scrape configuration, labels and network policies. I would also check Prometheus logs and resource utilisation. Once I identify whether the problem is instrumentation, service discovery, connectivity or Prometheus itself, I would restore scraping and validate that the metrics are appearing in Grafana and that alerts are functioning." ---

7. Grafana dashboard is green but users complain

Question

Everything looks green in Grafana, but customers say the application is slow. What do you do? **Very important L3 question.**

Spoken Answer

"I wouldn't assume the application is healthy just because the existing dashboard is green. First I would validate the actual user-facing symptoms—request latency, error rate and affected transactions. Then I would check whether we're missing application-level or dependency-level telemetry. I would look at P95/P99 rather than only averages, distributed traces, application logs and downstream dependencies such as databases or external APIs. I would also check whether the issue is regional, customer-specific or related to a recent deployment. If we identify a telemetry gap, I would address it after stabilising the incident so that the dashboard reflects the actual customer experience." ---

8. Alert fatigue

Question

Your team receives 100 alerts every day and most are false positives. What would you do?

Spoken Answer

"I would analyse the alert history rather than simply increasing thresholds. I would classify the alerts into actionable, informational and noise. For noisy alerts, I would determine whether the problem is an inappropriate threshold, evaluation window, duplicate alert or an alert that doesn't require human intervention. I would move appropriate alerts from paging to lower-severity notification, use grouping and deduplication, and align critical alerts with customer impact or SLO risk. After the changes, I would measure false-positive rate and verify that we're not suppressing genuine incidents. The objective is fewer unnecessary pages while maintaining incident detection." ---

9. Kubernetes pod keeps restarting

Question

A production pod is continuously restarting. How do you troubleshoot?

Spoken Answer

"I would first check the pod status and restart count using `kubectl get pods`. Then I would use `kubectl describe pod` to check events, probes, resource limits, scheduling and container state. I would also check the current and previous container logs using `kubectl logs` and `kubectl logs --previous`. I would specifically investigate OOMKilled, application crashes, failed health probes, configuration or secret issues, dependency failures and recent deployments. If it's impacting production, I would first mitigate according to the runbook—for example rollback or scaling—then investigate the underlying cause and implement a permanent fix." ---

10. Exit code 137

Question

What does exit code 137 mean?

Spoken Answer

"Exit code 137 generally means the process received SIGKILL, and in Kubernetes the most common cause is an OOMKilled container. I would confirm it using `kubectl describe pod` and check the container's memory requests and limits, actual memory consumption, restart count and node memory pressure. Increasing the memory limit can be a mitigation, but I wouldn't treat that as the permanent fix. I would investigate whether there is a memory leak, abnormal workload, inefficient application behaviour or incorrect resource sizing." ---

11. Kafka consumer lag is increasing

Question

Kafka consumer lag is continuously increasing. What do you check?

Spoken Answer

"First I would identify the scope—consumer group, topic and partition. Then I would check whether consumers are healthy, whether processing rate has decreased, whether there are consumer restarts or errors, and whether partition assignments are balanced. I would also compare producer throughput with consumer throughput. If producers suddenly increased traffic, the consumers may simply need additional capacity. Otherwise I would investigate broker health, disk, network and downstream dependencies. The key is to distinguish whether the root cause is increased producer volume, insufficient consumer capacity, partition imbalance, consumer failures or a slow downstream dependency." ---

12. P1 incident and stakeholder communication

Question

How do you communicate during a P1 incident?

Spoken Answer

"I keep communication concise and business-focused. I communicate what is impacted, when it started, the current scope, what we're doing to mitigate it and when the next update will be provided. For technical teams I provide detailed investigation information such as metrics, logs, traces and recent changes. For management or business stakeholders, I translate that into customer and business impact rather than using unnecessary technical terminology. I also maintain a clear incident timeline because it becomes important during the RCA." ---

13. You disagree with another senior engineer

Question

What would you do if another senior engineer disagrees with your proposed production solution?

Spoken Answer

"I would first understand their reasoning rather than treating it as a personal disagreement. I would compare both approaches using production evidence—customer impact, logs, metrics, risk and rollback options. If possible, I would choose the safest reversible approach and validate it. During a critical incident, the priority is restoring service, not proving who is right. If we still cannot agree and the decision has significant production risk, I would involve the incident lead or appropriate technical owner." ---

14. You make a mistake during a production incident

Question

Suppose you made a wrong change during a production incident. What would you do?

Spoken Answer

"I would immediately acknowledge the mistake rather than trying to hide it, assess the impact and reverse the change if it is safe to do so. I would communicate the updated situation to the incident lead and stakeholders. After stabilisation, I would include the event transparently in the incident timeline and RCA. Then I would look at why the mistake was possible—for example insufficient approval, unclear runbook or lack of automation—and introduce controls to prevent recurrence." ---

15. Business asks for faster deployment, but reliability is suffering

Question

Management wants faster releases, but your error budget is being consumed. What do you do?

Spoken Answer

"I would use the SLO and error budget as an objective way to have that discussion. If reliability is consuming the error budget faster than planned, I would explain the customer impact and reliability risk using data. I wouldn't simply stop releases indefinitely. I would look for ways to improve both reliability and delivery—for example automated testing, canary deployments, progressive delivery, rollback automation and better monitoring. The goal is to make the delivery process safer and faster rather than treating reliability and delivery as opposing objectives." ---

16. You inherit a system with poor monitoring

Question

You join a new project and there is almost no monitoring. What do you do in your first few weeks?

Spoken Answer

"I would first understand the critical business journeys and dependencies. Then I would establish baseline SLIs and define appropriate SLOs. I would implement the essential metrics around traffic, errors, latency and saturation, centralised logging and distributed tracing for critical flows. Then I would build dashboards around customer impact and create actionable alerts with ownership and runbooks. I would prioritise the highest-risk services first rather than trying to instrument everything at once. After implementation, I would validate the monitoring through controlled failure scenarios and continuously improve based on incidents." ---

17. No runbook exists for an incident

Question

You receive an alert but there is no runbook. What do you do?

Spoken Answer

"I wouldn't blindly make changes. I would first establish impact and scope, then use available telemetry—metrics, logs, traces and recent change information—to form a hypothesis. I would choose the safest reversible mitigation and involve the appropriate technical owners where required. Once the service is stable, I would document the investigation and recovery steps and convert them into a runbook. That turns the incident into an improvement for the next engineer." ---

18. Critical service is down but monitoring is also down

Question

What if your monitoring system itself fails during a production incident?

Spoken Answer

"I would recognise that we're dealing with two separate problems: loss of observability and potentially an application incident. I would use secondary telemetry such as application logs, cloud-native monitoring, infrastructure status, synthetic checks or service health endpoints to establish the application state. In parallel, I would restore the monitoring capability because without it our ability to operate the production system is degraded. Afterward, I would address monitoring as a critical dependency by looking at HA, redundancy and independent monitoring of the monitoring stack." ---

19. Executive asks: “When will it be fixed?”

Question

You don't know the exact ETA. What do you tell management?

Spoken Answer

"I wouldn't provide an artificial ETA just to give an answer. I would say something like: ‘The service is currently impacted, we've identified the affected component and the team is working on mitigation. Our immediate priority is restoration. We are evaluating rollback/failover as the recovery option, and I'll provide the next update in 15 minutes.’ I prefer giving a confirmed next-update time rather than making an unreliable promise about final resolution." ---

20. L3 leadership scenario

Question

What does leadership mean to you if you don't have direct reports?

Spoken Answer

"For me, technical leadership doesn't necessarily require direct reports. During production incidents, I take ownership of the problem, establish priorities, divide the investigation across team members and make sure communication is happening. For example, one engineer can investigate recent deployments, another can check infrastructure, while I correlate application logs, metrics, traces and dependencies. I make decisions based on customer impact and available evidence, and after recovery I make sure the RCA and preventive actions are completed."

L3 Universal Troubleshooting Framework

Impact → Scope → Triage → Metrics → Logs → Traces → Recent Changes → Dependencies → Isolate → Mitigate → Validate → RCA → Prevent

Core L3 themes: customer impact, SLOs, evidence-based diagnosis, safe mitigation, clear communication, ownership, and prevention.